Abstract

The tumor suppressor protein p53, encoded by the *TP53* gene, is among the most extensively studied proteins in cancer biology due to its central role in maintaining genomic integrity. In this project, we retrieved the human TP53 protein sequence (UniProt: P04637) and performed a comparative homology search using BLASTP against the non-redundant protein database. The analysis revealed near-complete sequence conservation between human and gorilla TP53 orthologs, with 100% query coverage, an E-value of 0.0, and ~99–100% pairwise identity. These findings underscore the extreme evolutionary constraint acting upon TP53 across mammalian lineages, reflecting its indispensable function in DNA repair, cell cycle arrest, and apoptosis. This work demonstrates how publicly available bioinformatics tools and databases can be leveraged to investigate protein evolution and functional conservation without requiring experimental laboratory infrastructure.

Comparative Analysis of Human TP53 Protein Using BLASTP: A Bioinformatics Approach to Studying Evolutionary Conservation

1. Introduction

Bioinformatics has fundamentally changed how we approach biological questions. Rather than beginning with a wet-lab experiment, researchers can now start with sequence data, using computational tools to generate hypotheses about protein function, structure, and evolutionary history. For students entering the field, learning to navigate these tools is not just a technical exercise—it is an introduction to how modern biological research actually operates.

The protein I chose to investigate is p53, the product of the *TP53* gene. Often called the "Guardian of the Genome," p53 is a transcription factor that responds to cellular stress by activating pathways that either halt the cell cycle to allow for DNA repair or trigger programmed cell death when damage is irreparable. Because of this critical role, mutations in *TP53* are found in over half of all human cancers, making it one of the most clinically significant proteins in molecular oncology.

What makes p53 particularly interesting from an evolutionary perspective is the expectation that proteins with such vital functions tend to be highly conserved across species. If a protein is essential for survival, natural selection generally disfavors mutations that alter its core structure or function. This project was designed to test that hypothesis computationally: by comparing the human TP53 sequence against a broad database of known proteins, I aimed to quantify how well this protein has been preserved across evolutionary time and to identify which species share the closest orthologous relationships with the human variant.

2. Objective

The primary objective of this project was to perform a comparative sequence analysis of human TP53 using BLASTP to identify homologous proteins across different organisms, evaluate the degree of sequence conservation, and interpret the biological significance of these findings in the context of protein function and evolutionary biology.

3. Tools and Databases Used

UniProt (https://www.uniprot.org/)
A comprehensive, manually curated protein sequence and annotation database maintained by the European Bioinformatics Institute. UniProt provides not only raw sequence data but also functional descriptions, domain architecture, post-translational modifications, and disease associations. For this project, UniProt served as the source of the verified human TP53 sequence.

NCBI BLASTP (https://blast.ncbi.nlm.nih.gov/)
The Basic Local Alignment Search Tool for proteins, developed by the National Center for Biotechnology Information. BLASTP compares an amino acid query sequence against a protein sequence database, identifying regions of local similarity and generating statistical scores that help distinguish meaningful biological relationships from random matches.

Non-Redundant Protein Sequences Database (nr)
A comprehensive NCBI database that aggregates protein sequences from multiple sources (GenBank, RefSeq, PDB, Swiss-Prot, PIR, PRF) while removing identical or nearly identical entries to reduce redundancy. This ensures that BLAST results represent diverse biological sources rather than being skewed by multiple submissions of the same sequence.

4. Methodology

The workflow was designed to mirror how a researcher might actually initiate a comparative protein study.

Step 1: Sequence Retrieval
I began by accessing the UniProt database and searching for the human p53 protein. I specifically selected the reviewed entry (indicated by a gold star in UniProt) with accession number P04637, which corresponds to the Cellular tumor antigen p53 from *Homo sapiens*. This entry contains a sequence of 393 amino acids. I noted that using a reviewed (Swiss-Prot) entry is important because these records have been manually curated by expert biologists, ensuring higher confidence in the sequence and annotation compared to automatically generated (TrEMBL) entries.

Step 2: FASTA Format Preparation
The sequence was downloaded in FASTA format, the standard text-based representation for biological sequences. The header line begins with the ">" symbol, followed by metadata such as the protein name, gene, organism, and accession number. The sequence itself appears on subsequent lines, with each line typically containing 60–80 amino acid characters. Understanding FASTA format is foundational because nearly every bioinformatics tool, including BLAST, accepts this as standard input.

Step 3: BLASTP Configuration
I navigated to the NCBI BLASTP interface and pasted the complete 393-amino-acid TP53 sequence into the query field. For the search database, I selected "Non-redundant protein sequences (nr)" to ensure broad taxonomic coverage. The algorithm parameters were left at

default settings, which use the BLOSUM62 substitution matrix and expect a threshold (E-value) of 10. These defaults are generally appropriate for detecting evolutionary relationships among orthologous proteins.

Step 4: Homology Search and Result Extraction
After submitting the query, BLASTP executed a heuristic search that identified statistically significant alignments between the human TP53 query and sequences in the nr database. I focused my analysis on the top hits, paying particular attention to the query coverage, percent identity, and E-value for each match.

5. Results

The BLASTP analysis returned highly significant results. The top-scoring alignments are summarized below:

Table

Parameter	Top Match (Human TP53)	Gorilla TP53 Ortholog
Query Coverage	100%	100%
Percent Identity	100%	~99–100%
E-value	0.0	0.0
Accession	P04637	Gorilla ortholog entries

The E-value of 0.0 indicates that the probability of observing such a high-scoring alignment by chance is effectively zero, confirming that these matches represent genuine biological homology rather than random sequence similarity. The 100% query coverage indicates that the entire length of the human TP53 protein aligns with the matched sequences, suggesting that no major insertions, deletions, or truncations have occurred in these orthologs.

Among non-human matches, the gorilla TP53 protein emerged as one of the closest homologs, exhibiting ~99–100% amino acid identity. Other primates and mammals also showed extremely high conservation, with identity percentages consistently above 95%.

6. Interpretation and Biological Significance

These results carry meaningful biological implications that extend beyond the numbers.

First, the near-perfect conservation between human and gorilla TP53 is consistent with what we know about evolutionary divergence. Humans and gorillas shared a common ancestor approximately 8–10 million years ago. For a protein to retain ~99% identity over that timescale implies that nearly every amino acid in the sequence is under strong purifying selection. Any mutation that alters the protein's structure or function is likely to be deleterious and therefore removed from the population over evolutionary time.

Second, this conservation pattern supports the functional importance of TP53. As the Guardian of the Genome, p53 operates at a critical decision point in the cell: it integrates signals from DNA damage, oxidative stress, and oncogene activation to determine whether a cell should arrest, repair, or die. Because this decision is so central to organismal survival, the protein's architecture—including its DNA-binding domain, tetramerization domain, and regulatory regions—has been finely tuned by evolution and locked in place.

Third, from a practical standpoint, high conservation across mammals makes TP53 an excellent candidate for comparative modeling and cross-species translational research. If a drug or therapeutic strategy targets a specific region of human p53, understanding how that region is conserved in model organisms (such as mice, which share ~78% identity) becomes essential for preclinical studies.

Finally, the observation that even distantly related mammals retain recognizable TP53 homology suggests that the origins of this tumor suppressor pathway predate mammalian diversification. This aligns with the broader understanding that the p53 protein family (including p63 and p73) arose early in vertebrate evolution and has since been co-opted into increasingly complex regulatory networks.

7. Conclusion

This project demonstrated that a straightforward bioinformatics workflow—retrieving a sequence from UniProt and querying it via BLASTP—can yield rich insights into protein evolution and functional biology. The human TP53 protein is extraordinarily conserved across mammals, with gorilla and human orthologs being nearly identical at the amino acid level. This conservation reflects the non-negotiable importance of p53 in genome maintenance and cellular homeostasis.

For me as a student, this project reinforced a key principle: bioinformatics tools are not black boxes that produce abstract outputs. When interpreted carefully, their results tell biological stories. The E-value, percent identity, and query coverage are not just statistics; they are quantitative measures of evolutionary pressure and functional constraint. Moving forward, I plan to extend this analysis by examining TP53 mutations documented in the COSMIC database and exploring how specific cancer-associated substitutions disrupt conserved residues—a natural next step that builds directly upon the foundation established here.

8. References

1. UniProt Consortium. (2023). *UniProt: the Universal Protein Knowledgebase in 2023*. Nucleic Acids Research, 51(D1), D523–D531. https://www.uniprot.org/
2. Altschul, S. F., Gish, W., Miller, W., Myers, E. W., & Lipman, D. J. (1990). *Basic Local Alignment Search Tool*. Journal of Molecular Biology, 215(3), 403–410.
3. NCBI Resource Coordinators. (2018). *Database Resources of the National Center for Biotechnology Information*. Nucleic Acids Research, 46(D1), D8–D13. https://blast.ncbi.nlm.nih.gov/

4. Vousden, K. H., & Prives, C. (2009). *Blinded by the Light: The Growing Complexity of p53*. Cell, 137(3), 413–431.
5. Joerger, A. C., & Fersht, A. R. (2008). *Structural Biology of the Tumor Suppressor p53*. Annual Review of Biochemistry, 77, 557–582.